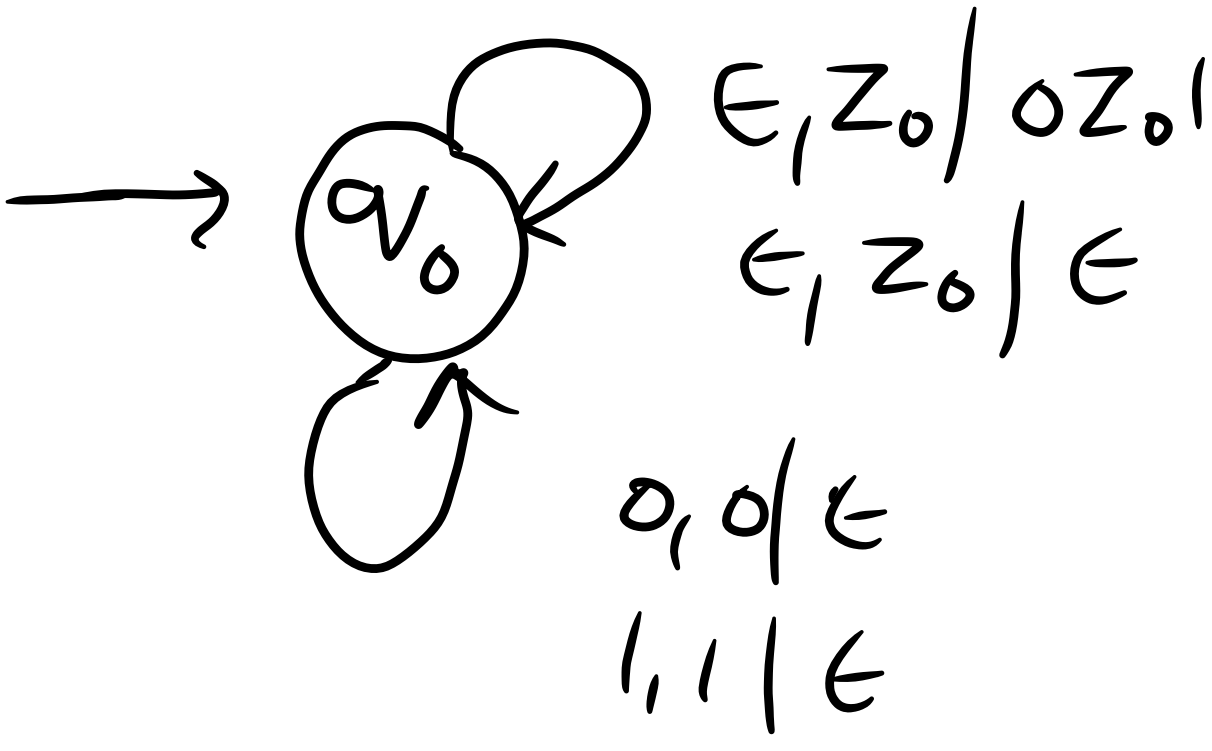
eg:- $L = \{0^n 1^n \mid n \geq 0\}$

cfg:-

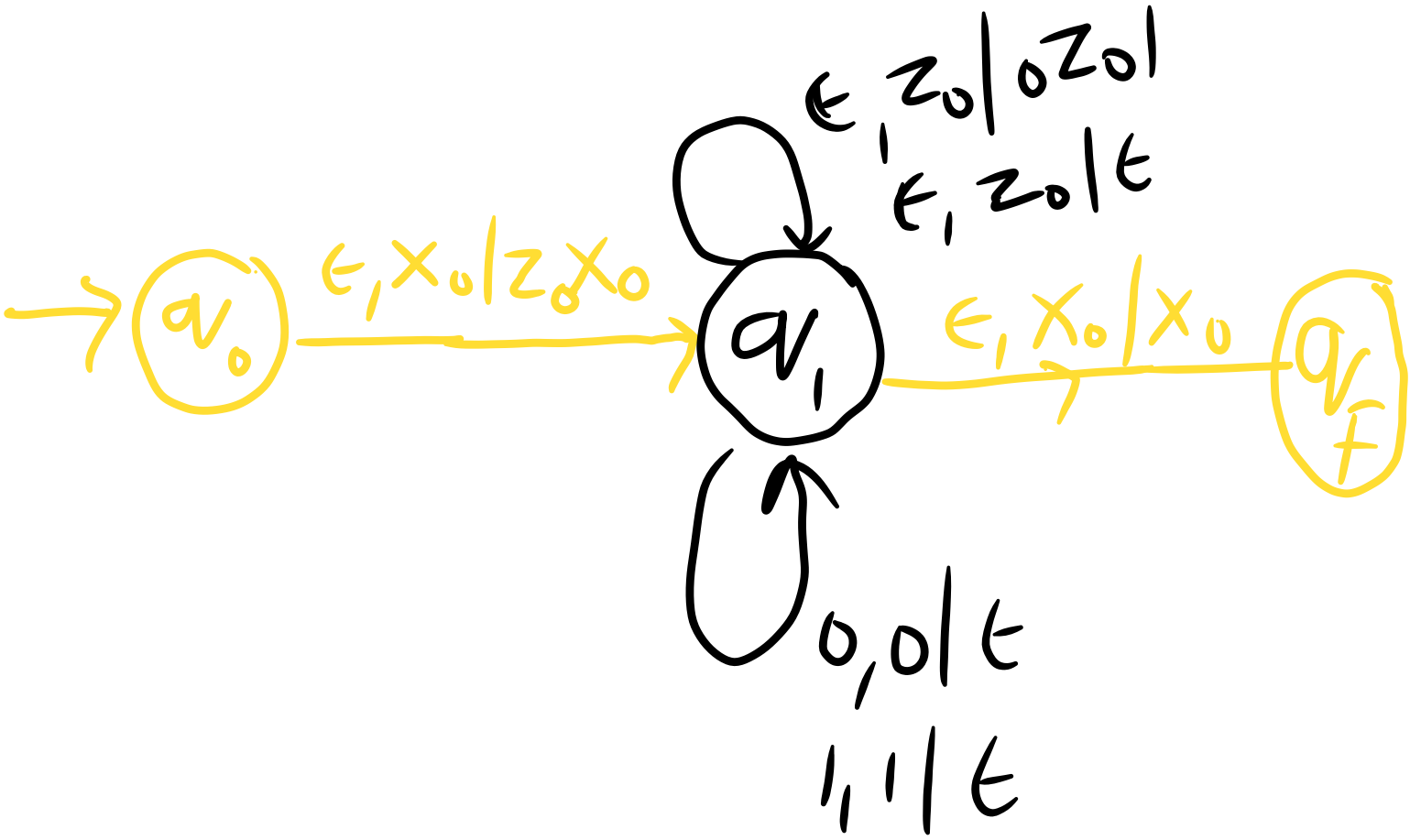
$S \rightarrow 0S1 \mid \epsilon$

PDA By Empty Stack



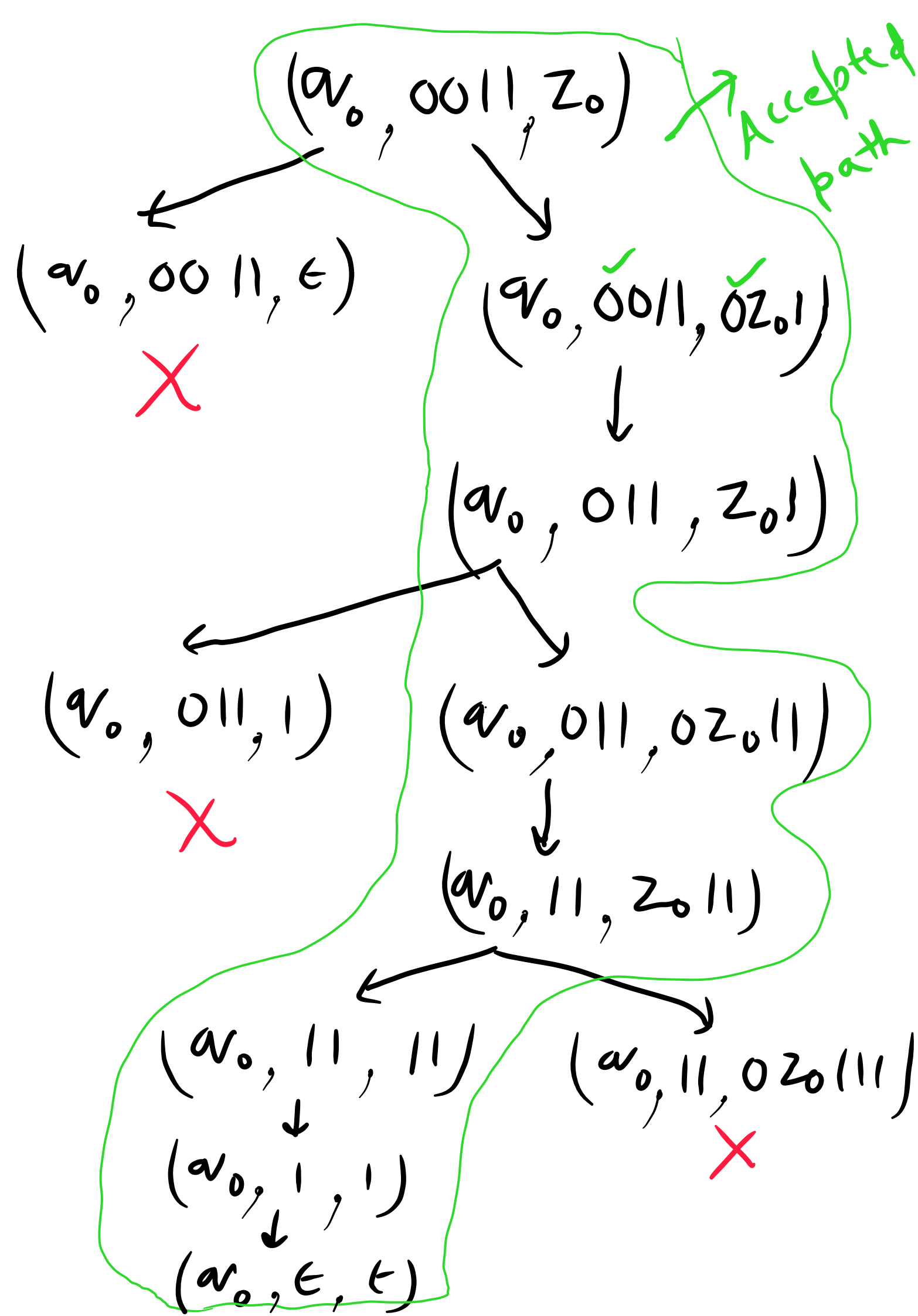
where $S = Z_0$

Final state PDA



String simulation :-

String :- 0011


$(q_0, 0011, Z_0)$
Accepted path
$(q_0, 0011, \epsilon)$
X
$(q_0, 0011, 0Z_01)$
$(q_0, 011, Z_01)$
$(q_0, 011, 1)$
X
$(q_0, 011, 0Z_011)$
$(q_0, 11, Z_011)$
$(q_0, 11, 11)$
$(q_0, 11, 0Z_0111)$
X
$(q_0, 1, 1)$
$(q_0, \epsilon, \epsilon)$

Context-Free Grammar Problems with Solutions

1. **Language:**
$$L = \{a^i b^j c^k \mid i = j \text{ or } j = k\}$$
CFG:
$$S \to S_1 \mid S_2$$
$$S_1 \to aS_1b \mid \epsilon$$
$$S_2 \to bS_2c \mid \epsilon$$

---

2. **Language:**
$$L = \{a^n b^m c^n \mid n \geq 1, m \geq 1\}$$
CFG:
$$S \to aSc \mid aBc$$
$$B \to bB \mid b$$

---

3. **Language:**
Palindromes where the middle symbol must be a
CFG:
$$S \to a \mid aSa \mid bSb$$

---

4. **Language:**
Strings over $\{a, b\}$ with no two consecutive a's
CFG:
$$S \to bS \mid aA \mid \epsilon$$
$$A \to bS \mid \epsilon$$

---

5. **Language:**
$$L = \{a^i b^j c^k \mid i + j = k\}$$
CFG:
$$S \to aSc \mid B$$
$$B \to bBc \mid \epsilon$$

---

6. **Language:**

$$L = \{a^i b^j c^k \mid i = j \text{ and } j \neq k\}$$

CFG:

$$S \to AB \mid AC$$
$$A \to aAb \mid \epsilon$$
$$B \to cB \mid c$$
$$C \to cC \mid \epsilon$$

(Note: One part generates $k > j$, the other $k < j$)

---

7. **Language:**
Strings where:

- a's and b's alternate
- start and end with the same symbol

CFG:

$$S \to a \mid b \mid aBa \mid bAb$$
$$A \to a \mid aBa$$
$$B \to b \mid bAb$$

---

8. **Language:**

$$L = \{a^n b^m \mid n = m + 2\}$$

CFG:

$$S \to aaA$$
$$A \to aAb \mid \epsilon$$